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	Width	Length
N-MOS	3 λ	2 λ
P-MOS	4 λ	2 λ

NOR

	Worst Case
VIL	0.528 V
VIH	4.755 V
NML	0.528 V
NMH	0.245 V
Rising Delay	1.0613 ns
Falling Delay	622.6 ps

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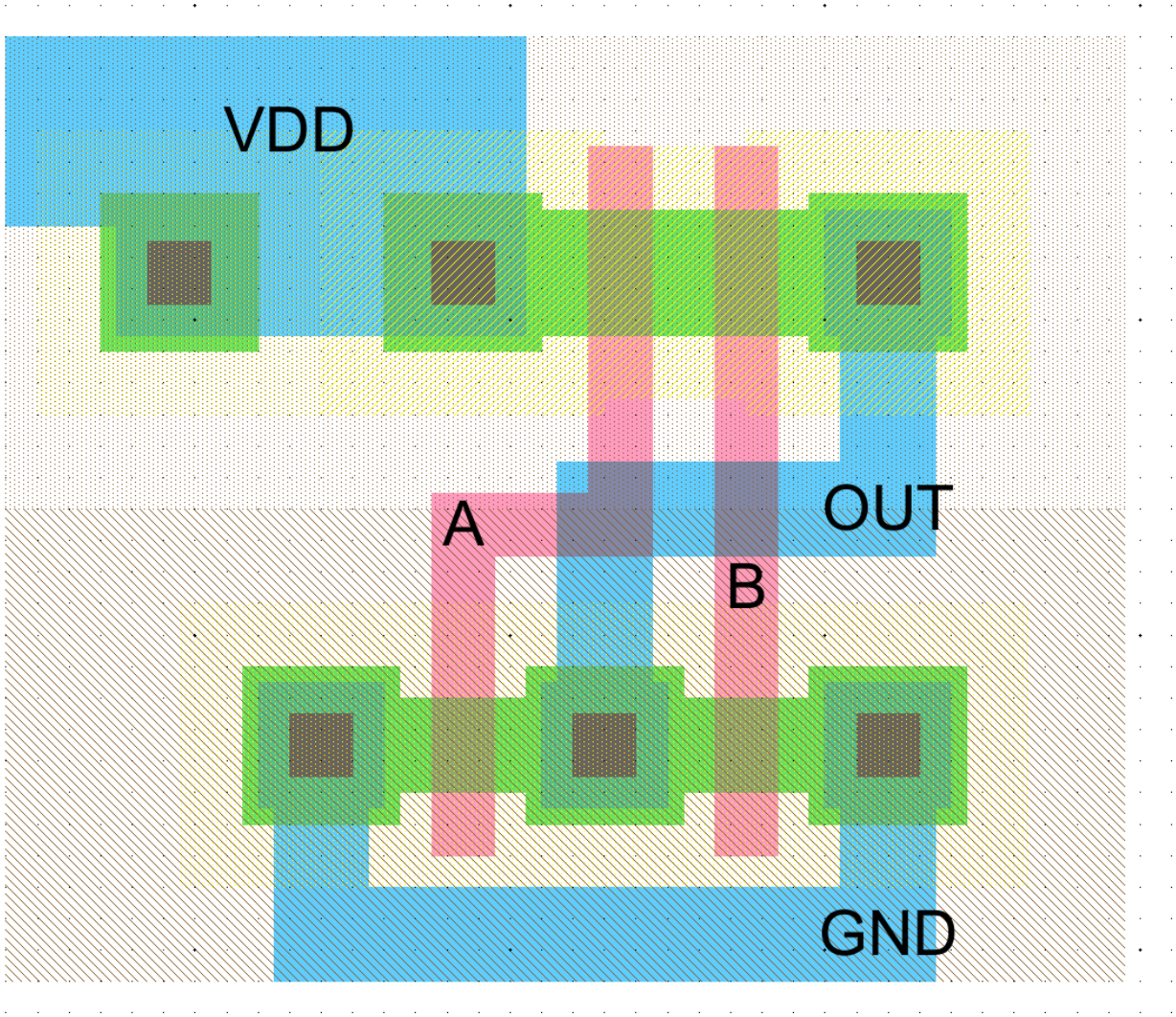

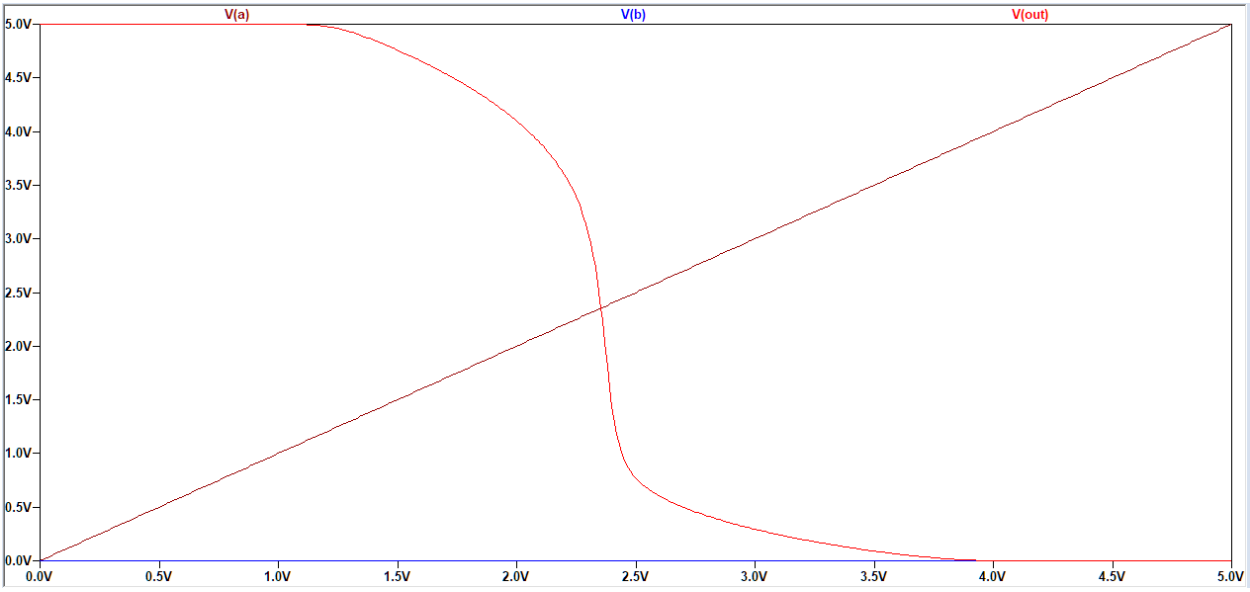

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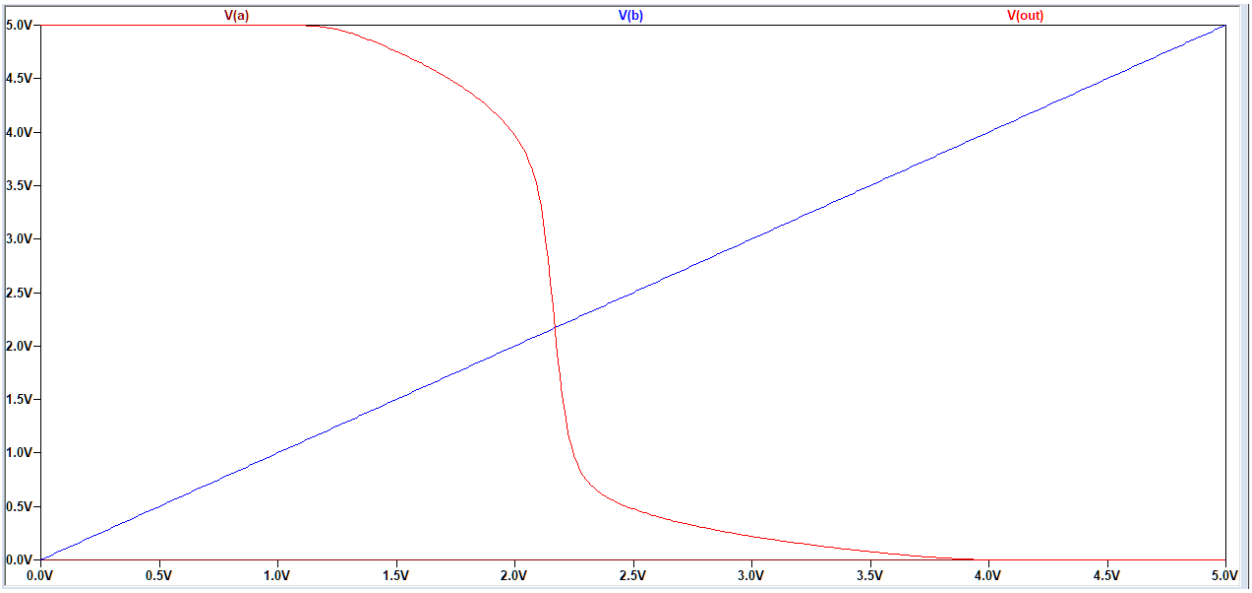
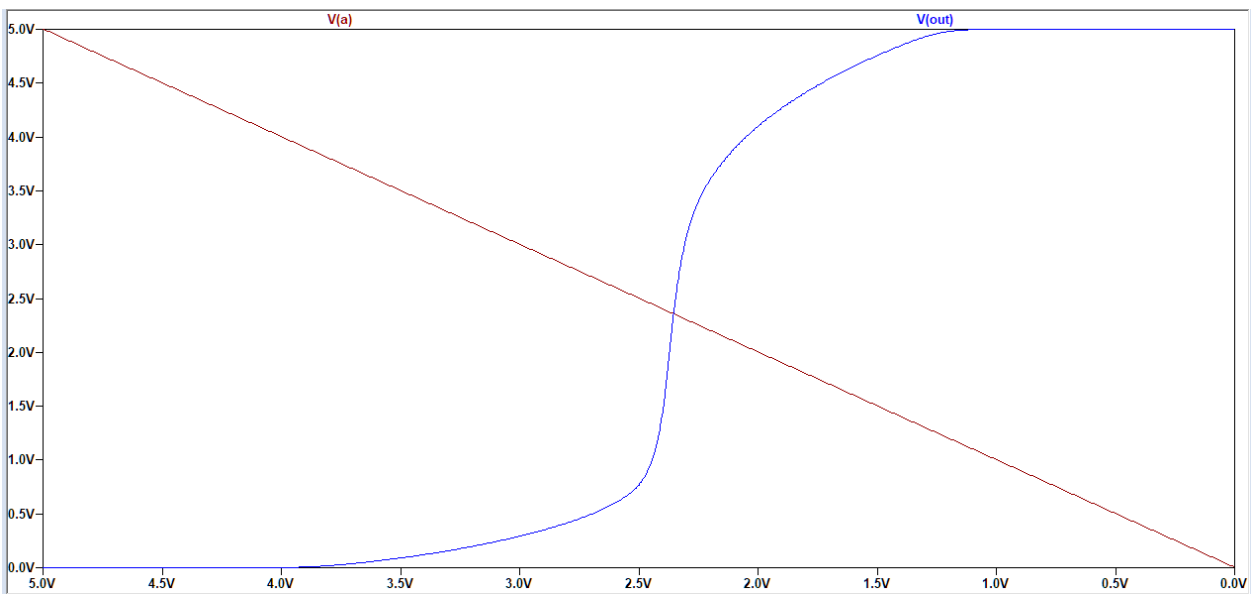
```
v_il: V(OUT) =0.534690729331 at 2.66035642916
v_ih: V(OUT) =4.72618990177 at 1.53004605405
nm_l: V_IL-0=0.534690729331
nm_h: 5-V_IH=0.273810098225
```




```
v_il: V(OUT) =0.528197800234 at 2.44017726367
v_ih: V(OUT) =4.7532609943 at 1.50028413146
nm_l: V_IL-0=0.528197800234
nm_h: 5-V_IH=0.246739005702
```




```
v_il: V(OUT) =0.537041967254 at 2.65799547755
v_ih: V(OUT) =4.73033750668 at 1.5259076
nm_l: V_IL-0=0.537041967254
nm_h: 5-V_IH=0.269662493324
```

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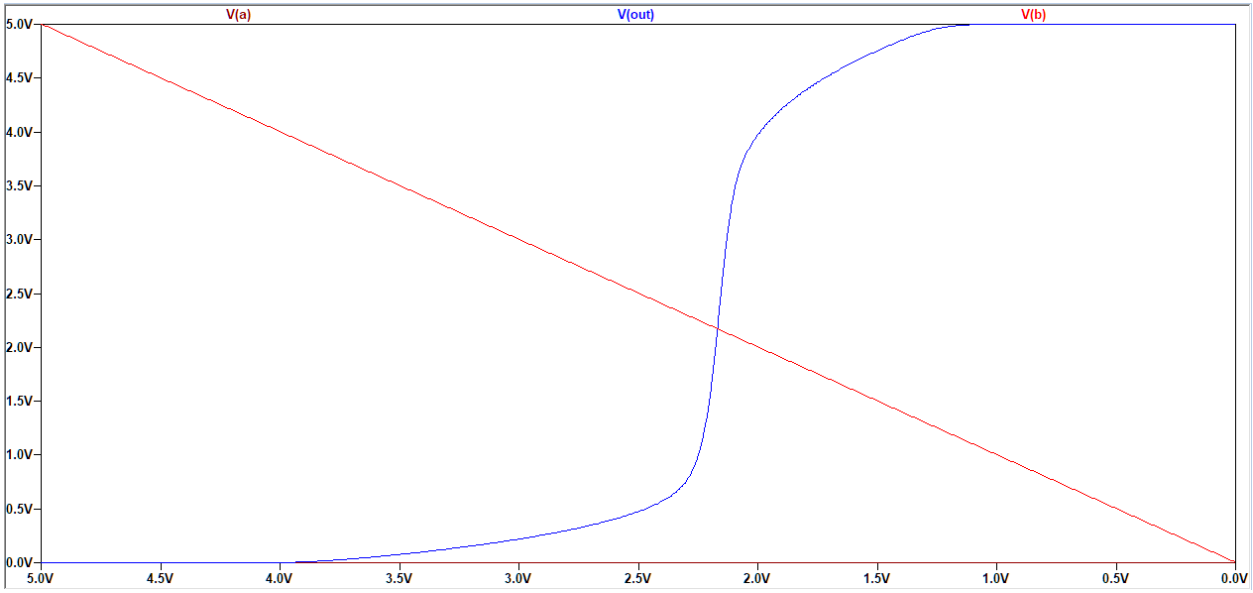


```
v_il: V(OUT) =0.531431350831 at 2.43691585882
v_ih: V(OUT) =4.75561099701 at 1.4979384
nm_l: V_IL-0=0.531431350831
nm_h: 5-V_IH=0.244389002991
```

Worst case (Smallest Noise Margins) occur when A is LOW and B moves LOW to HIGH

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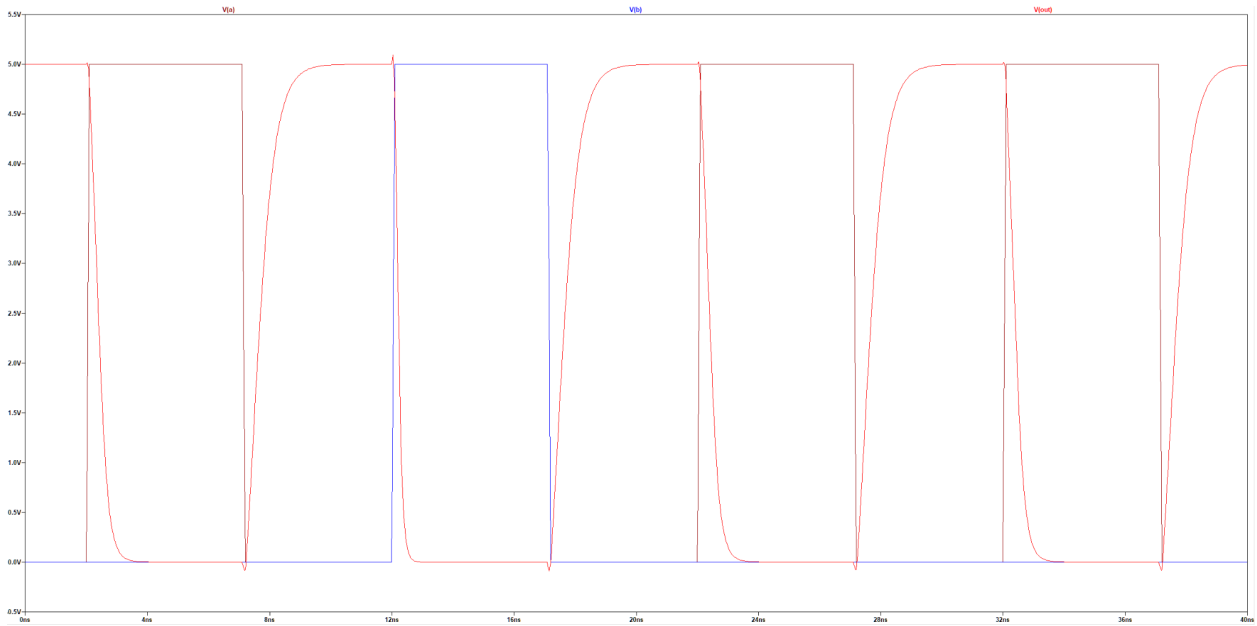
Transient Analysis of 80% Rise and fall time analysis




```
trise_1=1.06120665364e-09 FROM 7.32463094312e-09 TO 8.38583759675e-09
trise_2=1.0611787648e-09 FROM 1.72926027635e-08 TO 1.83537815283e-08
trise_3=1.06131245715e-09 FROM 2.73242198541e-08 TO 2.83855323112e-08
trise_4=1.06094467933e-09 FROM 3.73239275264e-08 TO 3.83848722057e-08
tfall_1=6.22329507651e-10 FROM 2.14038576682e-09 TO 2.76271527447e-09
tfall_2=2.77292628911e-10 FROM 1.21068641931e-08 TO 1.2384156822e-08
tfall_3=6.22629763413e-10 FROM 2.21406698994e-08 TO 2.27632996628e-08
tfall_4=6.21884224248e-10 FROM 3.21404012141e-08 TO 3.27622854384e-08
max_r12: max(trise_1, trise_2)=1.06120665364e-09
max_r34: max(trise_3, trise_4)=1.06131245715e-09
max_rise: max(max_r12, max_r34)=1.06131245715e-09
max_f12: max(tfall_1, tfall_2)=6.22329507651e-10
max_f34: max(tfall_3, tfall_4)=6.22629763413e-10
max_fall: max(max_f12, max_f34)=6.22629763413e-10
```

Worst Rise time is the third rise 1.0613 ns, occurring when A is LOW and B falls from HIGH to LOW
Worst Fall time is the third fall 622.6 ps, occurring when A is LOW and B rises from LOW to HIGH

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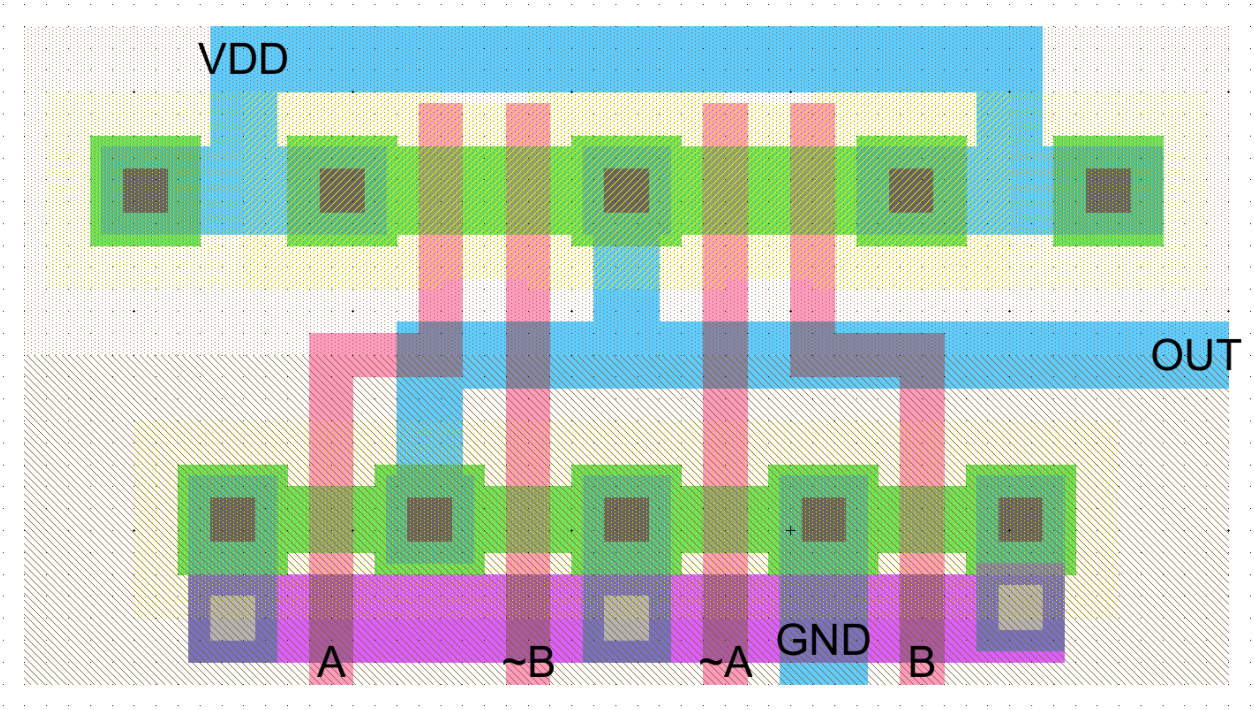
XOR



	Worst Case
VIL	0.574 V
VIH	4.747 V
NML	0.574 V
NMH	0.252 V
Rising Delay	6.436 ns
Falling Delay	1.309 ns

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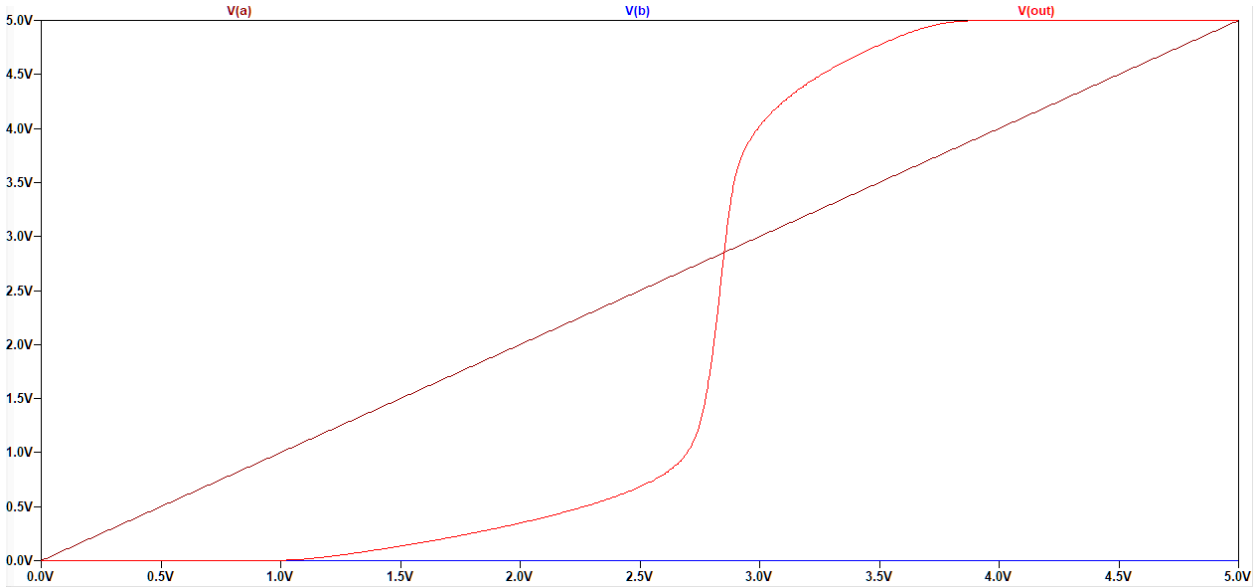
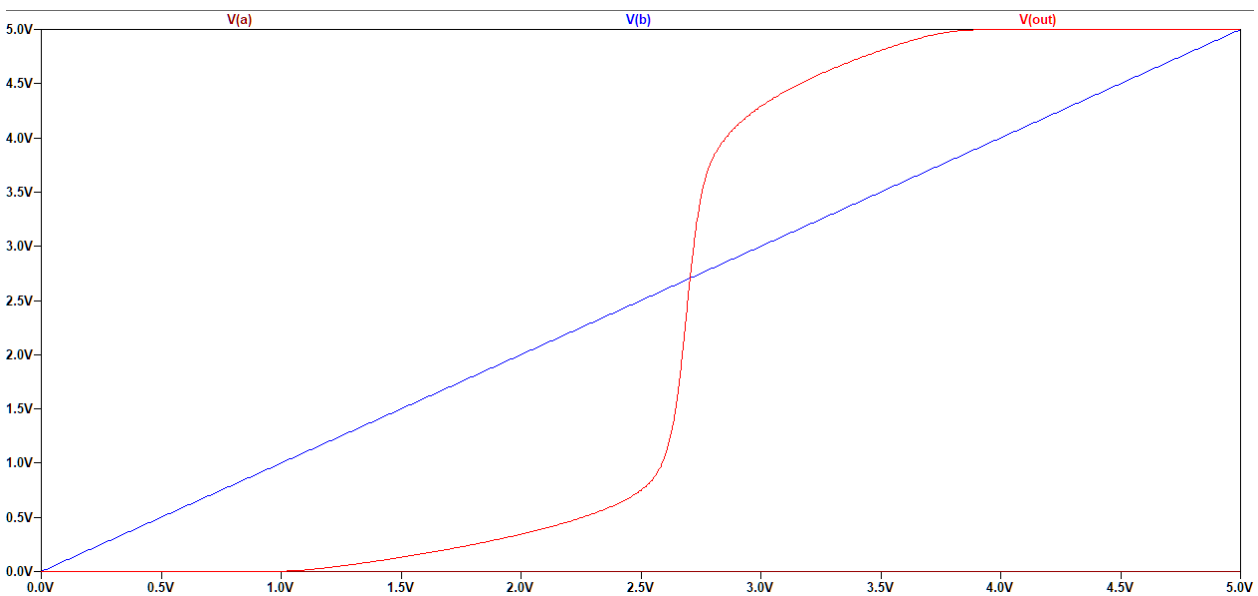



```
vil: V(OUT) =0.689494332113 at 2.50603685455
vih: V(OUT) =4.74715221539 at 3.4740924
nml: VIL-0=0.689494332113
nmh: 5-VIH=0.252847784615
```




```
vil: V(OUT) =0.613529121373 at 2.39109258039
vih: V(OUT) =4.57560978826 at 3.23502826667
nml: VIL-0=0.613529121373
nmh: 5-VIH=0.424390211741
```

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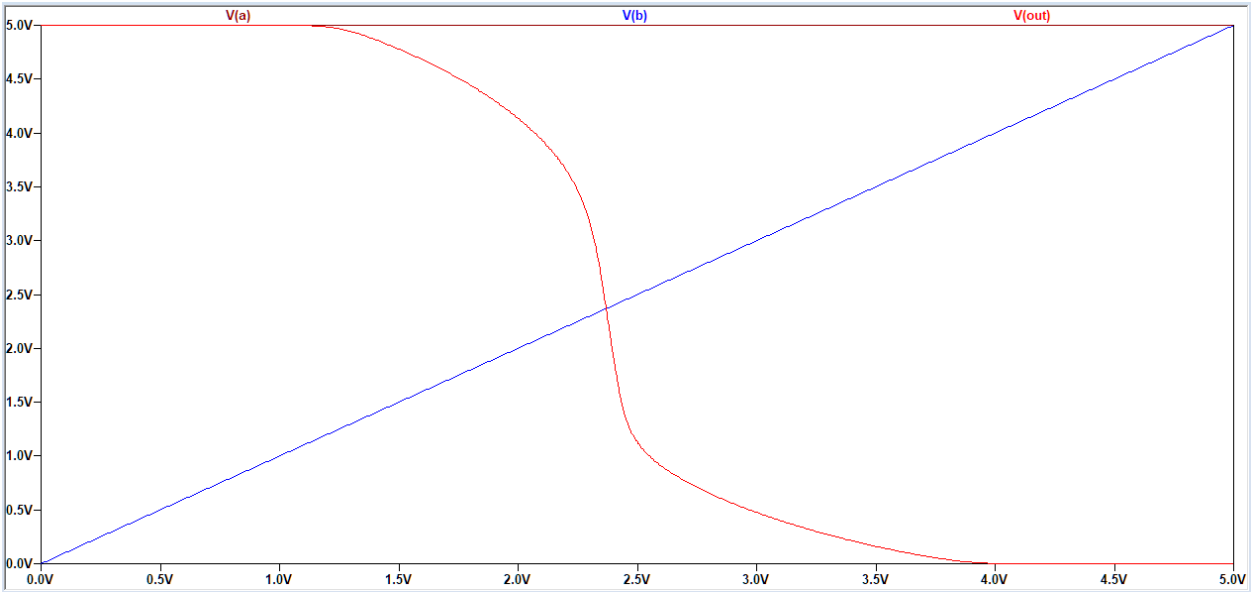
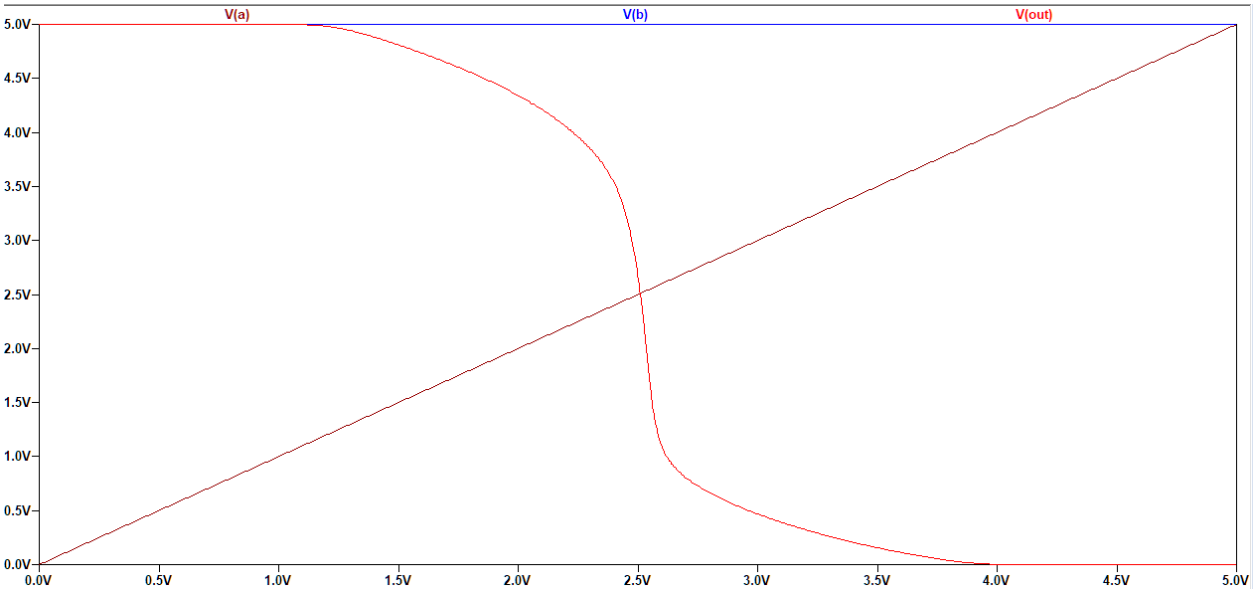



```
vil: V(OUT) =0.636461210077 at 2.81605301538
vih: V(OUT) =4.72210600262 at 1.5540076
nml: VIL-0=0.636461210077
nmh: 5-VIH=0.277893997383
```




```
vil: V(OUT) =0.574130887901 at 2.87907877647
vih: V(OUT) =4.51516153698 at 1.8390076
nml: VIL-0=0.574130887901
nmh: 5-VIH=0.48483846302
```

Worst case noise margin occurs when B is High and A goes from Low to High.

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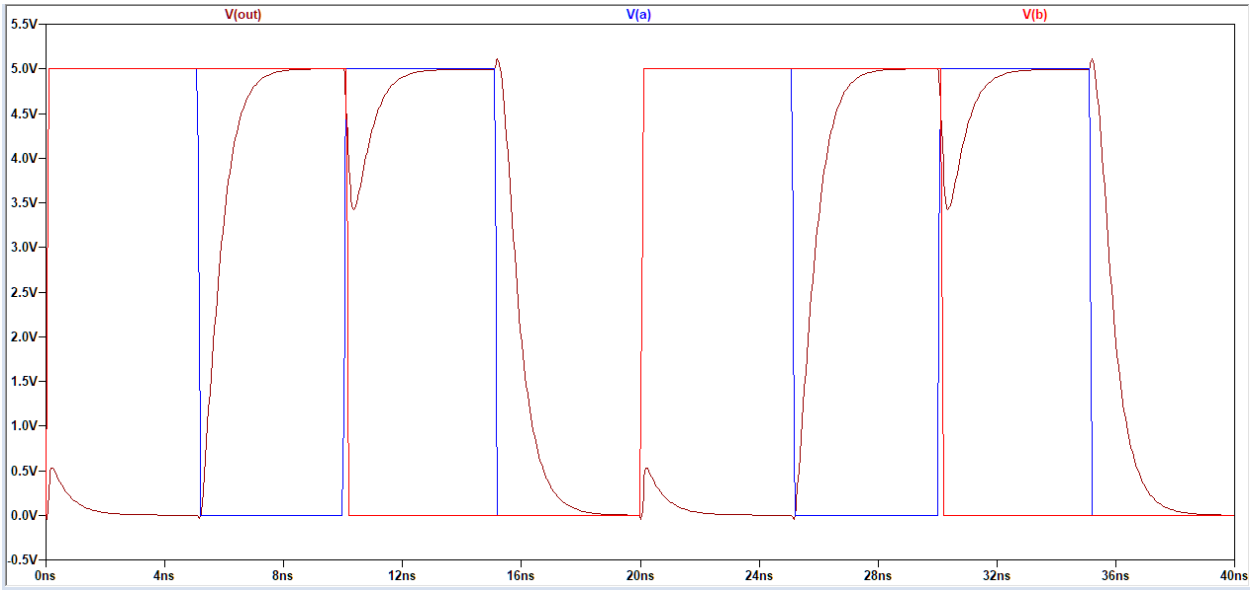
Transient Analysis of 80% Rise and fall time analysis




```
trise_1=6.43493841302e-09 FROM 1.50368953823e-10 TO 6.58530736684e-09
trise_2=5.81227694321e-09 FROM 5.33166009786e-09 TO 1.11439370411e-08
trise_3=6.43602979245e-09 FROM 2.01491725366e-08 TO 2.6585202329e-08
trise_4=5.81322453262e-09 FROM 2.53317967229e-08 TO 3.11450212555e-08
tfall_1=-9.8296200526e-09 FROM 1.01106514246e-08 TO 2.81031371973e-10
tfall_2=1.30955265912e-09 FROM 1.5439263665e-08 TO 1.67488163241e-08
tfall_3=-9.82761666093e-09 FROM 3.01102474647e-08 TO 2.02826308038e-08
tfall_4=1.30957348099e-09 FROM 3.54392173274e-08 TO 3.67487908084e-08
max_r12: max(trise_1, trise_2)=6.43493841302e-09
max_r34: max(trise_3, trise_4)=6.43602979245e-09
max_rise: max(max_r12, max_r34)=6.43602979245e-09
max_f12: max(tfall_1, tfall_2)=1.30955265912e-09
max_f34: max(tfall_3, tfall_4)=1.30957348099e-09
max_fall: max(max_f12, max_f34)=1.30957348099e-09
```

Worst Rise time is the third rise at 6.436 ns, occurring when B is HIGH and A falls from HIGH to LOW
Worst Fall time is the fourth fall at 1.309 ns, occurring when B are LOW and A falls from HIGH to LOW